

Simulation of Wildland Fire Fronts using the Level Set Method

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ABSTRACT

The prediction of wildland fire propagation is a critical component of modern fire management and safety protocols. Traditional Lagrangian methods, such as the Method of Lines (MOL), often fail or require complex redistancing algorithms when topological changes occur, such as fire fronts merging or fragmenting. This report details the implementation of an Eulerian Level Set Method to simulate wind-driven fire propagation. A two-dimensional numerical solver was developed from scratch to model the advection of a signed distance field. The solver was validated against multiple complex topological scenarios, demonstrating its inherent capability to seamlessly handle the merging of multiple fronts and the consumption of unburned fuel pockets without numerical instability.

1. Introduction

Wildland fire fronts propagate dynamically based on localized variables, including ambient wind velocity, topography, and fuel continuity. Historically, computational models have tracked these fronts using Lagrangian marker particles distributed along the fire boundary. While computationally light, this method breaks down during topological changes. When multiple fire fronts collide, the Lagrangian markers overlap and entangle, causing the numerical solver to crash unless computationally expensive marker-deletion algorithms are applied.

To resolve these topological limitations, this study employs the **Level Set Method**. Formulated as an Eulerian approach, the fire front is not explicitly tracked via points, but is instead implicitly defined as the zero-contour of a higher-dimensional scalar field, $\phi(x, y, t)$. By advecting this entire field over a fixed grid, topological changes such as merging fronts or the formation of unburned islands are handled naturally by the mathematics of the continuous field, requiring no explicit intervention by the solver.

2. Numerical Formulation

2.1. The Governing Equation

The core of the simulation is governed by the standard Level Set advection equation, a first-order hyperbolic partial differential equation (PDE):

$$\frac{\partial \phi}{\partial t} + S|\nabla \phi| = 0 \quad (1)$$

where ϕ is the signed distance function (with $\phi < 0$ representing burned areas and $\phi > 0$ representing unburned fuel), and S is the normal speed of the fire front.

2.2. Empirical Speed Models

The solver utilizes two distinct kinematic models for the rate of spread (S) depending on the simulated scenario. Both

rely on the base spread rate in zero wind (r_0) and the ambient wind vector ($\vec{V}$).

For the merging fire scenarios, the standard linear relationship was utilized:

$$S = r_0(1 + c_f(\vec{V} \cdot \hat{n})) \quad (2)$$

where c_f is a fuel-specific wind coefficient and $\hat{n}$ is the outward-pointing unit normal vector, defined as $\hat{n} = \frac{\nabla \phi}{|\nabla \phi|}$.

For the benchmark simulation, a more robust piecewise trigonometric model was implemented to accurately capture the differing propagation rates at the head, flanks, and rear of the fire. Let θ be the angle between the wind vector $\vec{V}$ and the normal vector $\hat{n}$, calculated via $\cos \theta = \frac{\vec{V} \cdot \hat{n}}{|\vec{V}|}$. The speed is defined as:

$$S = \begin{cases} r_0 \left(1 + c_f \sqrt{|\vec{V}| \cos^{1.5} \theta} \right) & \text{if } |\theta| < \frac{\pi}{2} \\ r_0(a + (1 - a) \sin \theta) & \text{if } |\theta| \geq \frac{\pi}{2} \end{cases} \quad (3)$$

where $a = 0.5$ dictates the backing rate against the wind.

2.3. Discretization and The Upwind Scheme

To maintain numerical stability when solving hyperbolic PDEs, information must propagate in the direction of the characteristics. To achieve this, a **First-Order Upwind Finite Difference Scheme** was implemented for the spatial derivatives.

The characteristic velocities in the x and y directions are defined as $U_x = Sn_x$ and $U_y = Sn_y$. The spatial derivative in the x -direction is approximated as:

$$\frac{\partial \phi}{\partial x} \approx \begin{cases} \frac{\phi_{i,j} - \phi_{i-1,j}}{\Delta x} & \text{if } U_x > 0 \\ \frac{\phi_{i+1,j} - \phi_{i,j}}{\Delta x} & \text{if } U_x \leq 0 \end{cases} \quad (4)$$

An identical logic is applied to the y -direction. Time integration was handled using an explicit Euler stepping method.

2.4. Simulation Parameters

All simulations were executed on a continuous domain of 100x100 meters. The grid was discretized into $N_x = 40$ and

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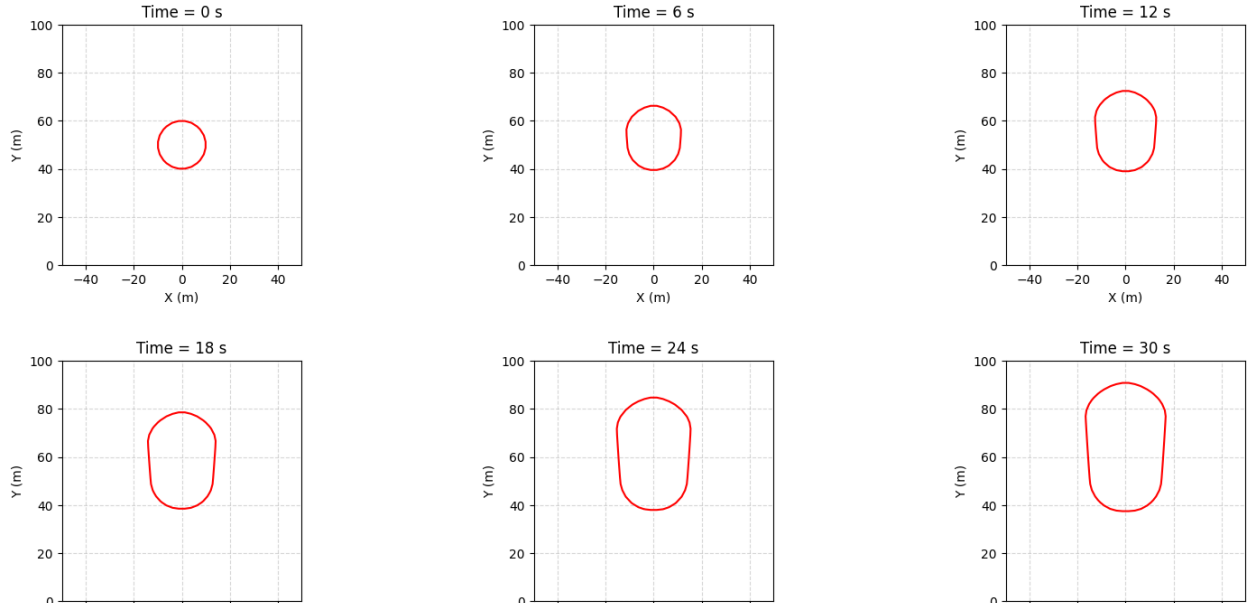


Figure 1: Snapshots of a single circular fire growing every 6 seconds.

$N_y = 40$ nodes, yielding a spatial resolution of $\Delta x = \Delta y = 2.56$ meters. The time step was set to $\Delta t = 0.1$ seconds. The physical constants were defined as $r_0 = 0.165$ m/s, $c_f = 3.24$, and a uniform ambient wind vector $\vec{V} = (0, 3)$ m/s.

3. Verification and Results

The numerical solver was subjected to four topological scenarios to verify its stability and accuracy against established fluid dynamic benchmarks.

3.1. The Benchmark Fire

A singular 10-meter radius circular fire was initialized at $(x = 0, y = 50)$. Under the influence of the positive y -direction wind, the head of the fire accelerated forward.

As demonstrated in Figure 1 and Figure 2, the piecewise trigonometric model (Equation 3) successfully anchored the base of the fire near $y = 40$ using the regulated backing rate ($a = 0.5$), preventing un-physical negative regression while the flanks exhibited lateral expansion.

3.2. Multiple Front Mergers

The solver's topological robustness was tested by simulating colliding fronts using the standard linear speed model (Equation 2). Figure 3 depicts a straight-line fire front propagating in the positive y -direction intersecting a circular spot fire.

Through the Level Set implicit representation ($\phi = \min(\phi_{line}, \phi_{spot})$), the two distinct geometries transitioned seamlessly into a continuous boundary without the need for localized grid manipulation. A subsequent three-front simulation (Figure 4) successfully demonstrated lateral merging into a uniform scalloped front.

To further stress-test the spatial logic, a three-front merger was initialized utilizing three symmetrical 10-meter circular fires placed evenly along the X -axis. Under the influence of the ambient wind, the heads of the fires accelerated in the positive y -direction, while the flanks expanded laterally at the base spread rate ($r_0 = 0.165$ m/s) until they intersected.

The Level Set algorithm effortlessly handled the simultaneous collision of these three boundaries. Instead of suffering from overlapping grid errors or requiring complex redistancing algorithms, the implicit $\phi = \min(\phi_1, \phi_2, \phi_3)$ representation naturally resolved the topological changes. The three distinct circles seamlessly merged into a single, continuous scalloped front that continued to propagate uniformly.

3.3. Unburned Fuel Pockets

The formation of unburned fuel islands poses a severe risk in real-world firefighting. Two spot fires and a line fire were initialized to enclose a central region.

Figure 5 captures the topological closure of the fronts, forming an isolated positive ϕ region (the island). The solver accurately advected the internal boundaries inward until the zero-contour collapsed entirely, signaling the complete consumption of the fuel pocket by 30 seconds.

4. Convergence Analysis

To verify the solver's consistency, a grid convergence study was conducted on the benchmark simulation. The domain resolution was iteratively increased from 30x30 to 60x60 nodes. The maximum Y -coordinate of the fire head at the final time step ($t = 30$ s) was recorded to measure truncation error against grid refinement.

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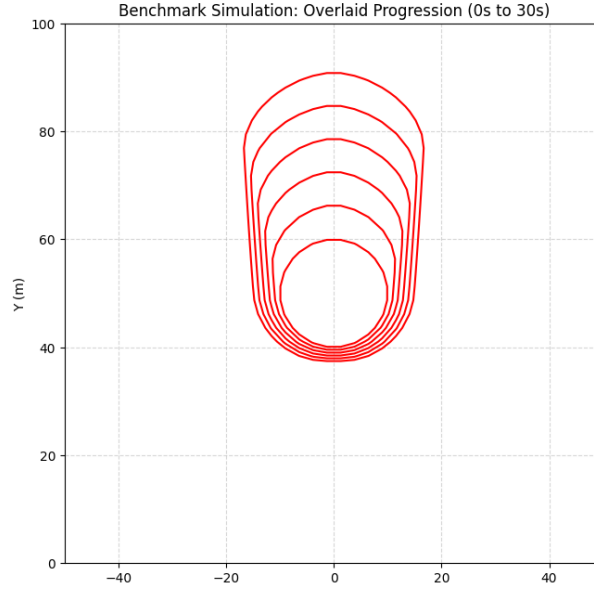


Figure 2: All time steps overlaid to show the total path of the benchmark fire.

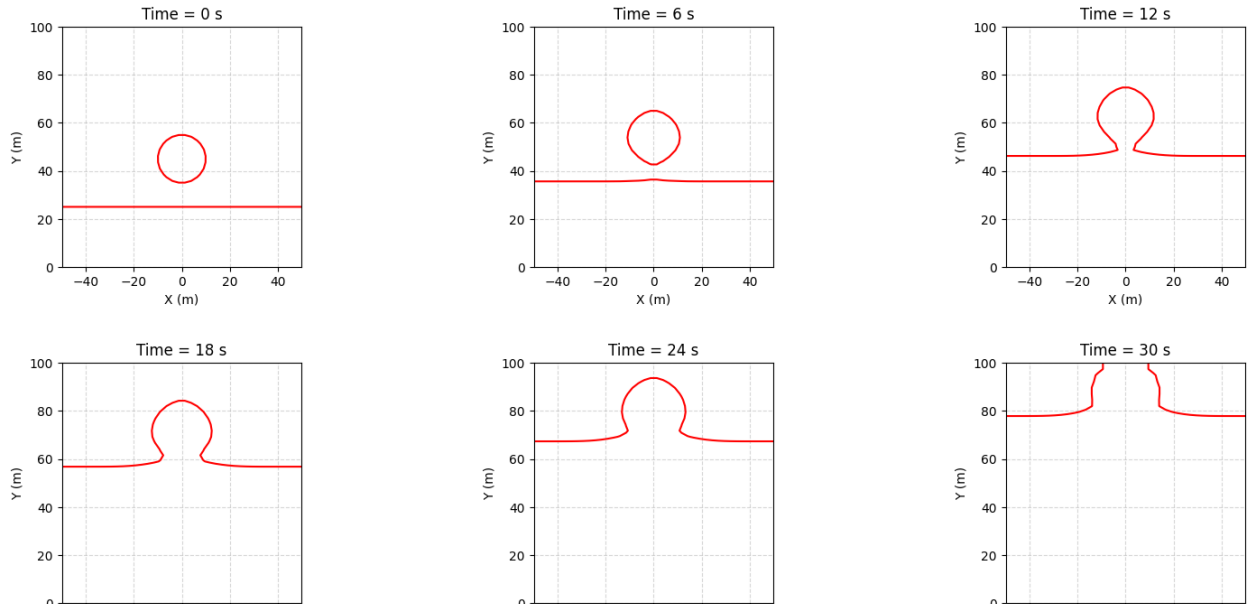


Figure 3: A straight line fire crashing into a circular spot fire.

As shown in Table 1, tracking the final y -position of the fire head revealed that the truncation error decreased proportionally with Δx (error between resolution decreasing from 1.16 to 0.64 to 0.37). This confirmed that the first-order upwind scheme was correctly implemented, providing a stable, monotonically decreasing error bounded by the grid resolution.

5. Numerical Anomalies and Fixes

During development, specific mathematical singularities arose that required explicit handling within the numerical scheme:

1. **Zero-Gradient Singularity:** When calculating the normal vector $\hat{n} = \nabla\phi/|\nabla\phi|$, perfectly flat fronts (such as the line fire in Figure 3) produced a gradient magnitude of exactly zero, leading to division-by-zero fatal errors (NaN). This was resolved by adding a

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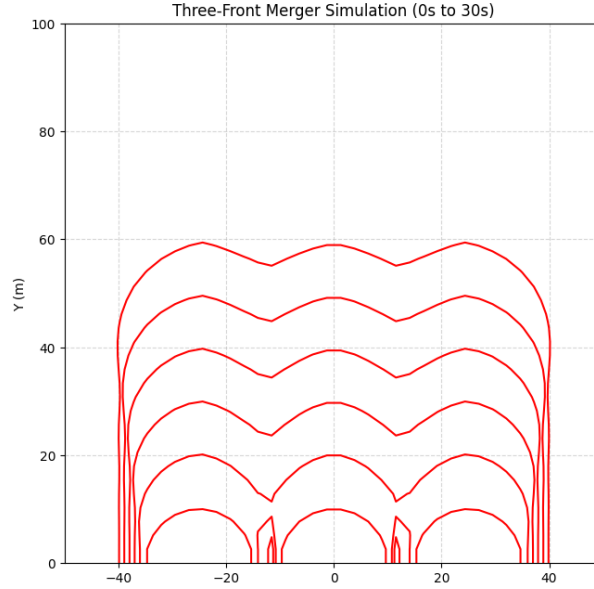


Figure 4: Three spot fires merging into a single scalloped front.

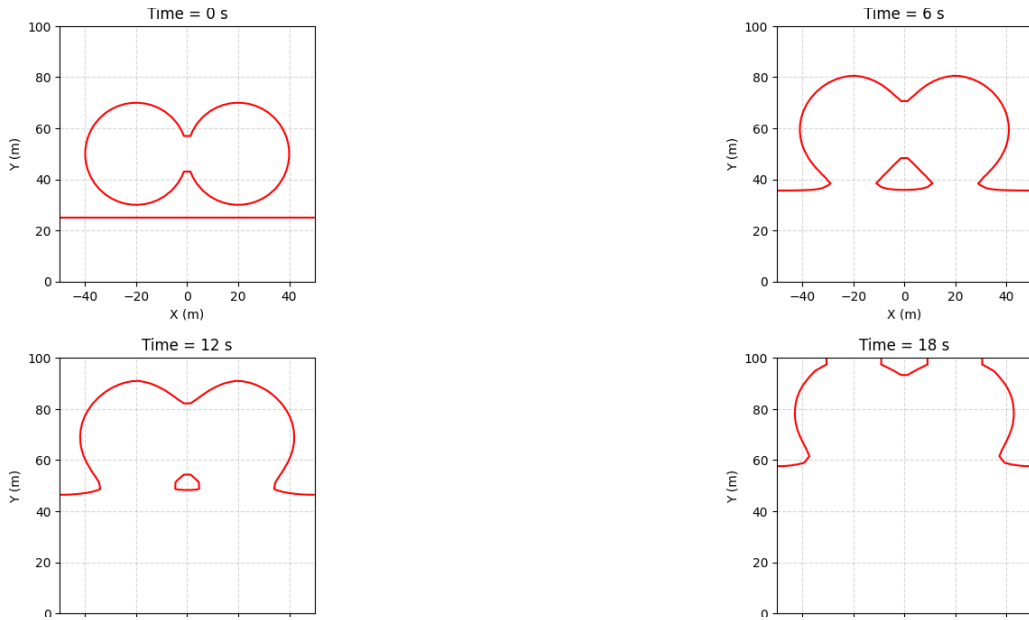


Figure 5: A pocket of unburned fuel gets trapped and fully consumed by $t = 30$ seconds.

microscopic offset $\epsilon = 1 \times 10^{-10}$ to the magnitude calculation, maintaining computational stability without altering the macroscopic physics.

2. **Negative Spread Rates in Linear Models:** The raw linear kinematic equation (Equation 2) can allow ambient wind vectors to overpower the base spread rate at the rear flank, mathematically forcing the fire to "unburn" fuel. Substituting the piecewise trigonometric function (Equation 3) for the benchmark case effectively bounded the solver, ensuring a consistent non-negative backing rate regardless of wind magnitude.

6. Conclusion

This study confirms that the Eulerian Level Set Method is a highly effective computational framework for wildland fire propagation. Unlike Lagrangian formulations, the advection of a continuous scalar field implicitly handles complex topological changes such as merging fronts and fuel island consumption without supplementary algorithmic intervention. The developed Python solver, utilizing a first-order upwind scheme and specialized kinematic models,

Table 1

Convergence of the fire head position ($t = 30$ s) across varying grid resolutions.

Nodes ($N_x \times N_y$)	Cell Size Δx (m)	Max Fire Head Y (m)
30×30	3.45	89.6882
40×40	2.56	90.8489
50×50	2.04	91.4885
60×60	1.69	91.8569

proved stable, accurate, and capable of reproducing the critical topological behaviors documented in benchmark fluid dynamics literature.

References

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